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From Cloud LLMs to Embodied AI: The Next Hardware Investment Frontier and Macro Regime Shift

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Introduction

In 2011, Marc Andreessen declared that "software is eating the world," foreseeing a transformation where software-driven companies would disrupt and dominate traditional industries. Over the next decade, that prediction came true. Cloud computing, mobile technology, and digital services reshaped global equity markets, driving massive gains in the S&P 500 and MSCI World Index. Technology firms, led by the so-called "Magnificent 7," surged to dominate market weightings, while sectors like manufacturing and energy saw their influence wane. Code became the world's most valuable commodity, lifting the Magnificent 7 Price-to-Sales ratio from under 2 in 2013 to nearly 8 at its peak in 2024.

Now, in 2025, we stand at the brink of a new era—one that may prove even more transformative. The rise of artificial intelligence marks a turning point not only in how software is built and deployed, but in how machines interact with the physical world. As Elon Musk put it, "AI is the most profound technology that humanity will ever develop… more profound than electricity or fire." If software ate the world, AI is poised to rebuild it—rewiring industries, infrastructure, and even investor expectations from the ground up.

The convergence of AI and hardware advancements is unfolding against the backdrop of a profound macro regime shift, driven by simultaneous changes in fiscal, monetary, and innovative forces. In the era of software dominance, the Federal Reserve's zero interest rate policy (ZIRP) fueled liquidity, inflating valuations across tech sectors. The COVID-19 crisis acted as a catalyst, unleashing an unprecedented wave of monetary and fiscal interventions to sustain liquidity. By 2022, with inflation surging and the Fed hiking rates aggressively, technology stocks—burdened by lofty valuations—faced pressure as the "higher for longer" narrative took hold, only for ChatGPT's debut to temporarily allay fears for high-multiple names. Now, a new administration is poised to tighten fiscal spending, raise tariffs and curb deficits, coinciding with the explosive rise of AI—a transformative innovation democratizing intelligence on an unprecedented scale—marking a pivotal shift in the economic and technological landscape.

AI is already democratizing coding and decision-making, accelerating the pace of innovation and eroding traditional moats. The Magnificent 7—once clear leaders in AI adoption—now face escalating capital expenditures and increased competition as they race toward artificial general intelligence (AGI). The sheer scale of infrastructure needed for this evolution—from data centers and semiconductors to energy and cooling—has ignited a global arms race between the U.S. and China. While some question whether this spending will pay off, the capital is already being deployed.

But what makes this moment different from previous tech cycles is that AI is beginning to bridge the digital and physical worlds. The post-software era will not be defined solely by algorithms and cloud platforms, but by the rise of intelligent machines: robots, self-driving vehicles, AI-powered smartphones, and physical agents operating in real time. This shift will reallocate market leadership toward sectors that can build, power, and move these embodied systems—especially in hardware, energy, semiconductors, and manufacturing.

The escalating power demands of AI underscore a seismic shift looming on the horizon, as electricity consumption is poised to skyrocket in the coming years. A Bloomberg Intelligence report, cited by Sprott, projects that data center electricity use could surge between fourfold and tenfold by 2030 under an aggressive AI growth scenario. At the upper limit, this would see data centers accounting for roughly 17% of U.S. power consumption by 2030. Though this represents a high-end estimate, it underscores the staggering potential scale of change if AI adoption races forward unchecked, amplified by a departure from the modest power needs of the recent software era and compounded by a decelerating China

As investors prepare for the next decade, understanding this transition is critical. The S&P 500 of 2030 will look very different from that of today—not just because of software innovation, but because AI is becoming embedded in the real world. We are entering a hardware renaissance, where code must be physically delivered through systems capable of sensing, reasoning, and acting.

This paper is the first in a series I'm writing at 22V Research to track the thematic macro impact of AI on global markets. The series begins with a deep dive into AI's hardware and infrastructure needs—an especially timely topic as doubts linger about the pace of physical buildout since DeepSeek. Each report will include macro data, company insights, and consultant research to help investors do their own work around each theme.

Eventually, this work will culminate in a thematic index of AI-exposed companies—grounded in fundamentals and corporate commentary—to help track these evolving narratives in real time. By combining this with an analysis of fiscal and geopolitical shifts under the new U.S. administration, the goal is to offer a forward-looking framework for what the new AI-powered S&P 500 might look like five years from now. If Andreessen's software thesis served as the roadmap for the last decade, this series aims to serve as a guide to the next: one where AI not only thinks but moves—and where the machines that carry that intelligence become the investment frontier.

From Cloud LLMs to Embodied AI: The Next Hardware Investment Frontier

Executive Summary

Artificial intelligence is entering a new phase, transitioning from software-based large language models (LLMs) running in the cloud to **embodied AI** integrated into hardware systems. This shift means AI will move beyond chatbots and cloud APIs into **physical agents** – from robots on factory floors to smart devices in our pockets and autonomous vehicles on roads. The significance of this transition is immense: it marries the cognitive power of AI with real-world action, promising leaps in productivity and automation across industries. Major analysts forecast trillions in economic value creation from AI's expansion – McKinsey, for example, estimates about *$4.4 trillion* in annual productivity potential from AI applications in the long term (AI in the workplace: A report for 2025 | McKinsey). As AI ventures into the physical world, it's poised to drive the next big tech cycle, similar to past shifts like the PC, internet, and mobile revolutions.

For investors, this emerging hardware-centric AI cycle presents substantial opportunities over the next 1–3 years. The forthcoming boom will focus on building and supplying the essential components of the AI era—advanced chips, sensors, robotics, and infrastructure—to enable large-scale AI deployment. Early indicators of this trend include significant capital expenditures by tech giants for AI data center construction and unprecedented demand for AI chips and components. In his GTC 2025 keynote, NVIDIA CEO Jensen Huang projected that data center capital expenditures could exceed $1 trillion by 2028, emphasizing the magnitude of the investment on the horizon. In essence, the AI boom's focus is expanding: following an initial phase centered on software (algorithms and models), the next growth stage will be hardware-driven, as companies strive to equip the world with embodied, intelligent systems. Investors should prepare for a surge in spending on the AI infrastructure and devices that will facilitate this transformation

Key takeaway: The advent of embodied AI represents an inflection point where investment is shifting toward hardware enablers of AI. This report outlines the evolution driving this shift, the critical hardware components needed, and the public companies (primarily U.S.-based, with notable players in Europe and Asia) best positioned to benefit. It also highlights the **scale of the opportunity** – measured in hundreds of trillions of dollars. The coming AI-driven growth cycle could see hardware spending outpace software, catalyzing growth for chipmakers, device manufacturers, and infrastructure providers powering the new era of AI.

The Evolution of AI: From Software to Hardware

Current State – Cloud AI with LLMs: Until now, most AI innovation has been software-centric. Large language models like GPT-4 have been trained and hosted in cloud data centers, accessible via APIs or web interfaces. Enterprises have so far adopted AI by integrating cloud-based models into their software workflows (for example, using AI for customer service chatbots or code generation). This cloud-first paradigm has made AI widely available but keeps the "intelligence" largely confined to server racks. AI's impact has thus been significant in digital realms – analyzing text, images, and data – yet mostly **virtual and disembodied**.

The Next Phase – AI Embodied in the Physical World: We are now at the cusp of AI moving from the cloud into **physical devices and machines** that operate in the real world. These embodied AI systems range from **enterprise AI agents** (smart co-workers that might autonomously handle tasks), to **personal AI devices** (like intelligent smartphones, home robots or wearables), to advanced **humanoid robots**, and self-driving **autonomous vehicles**. Unlike static cloud AI, embodied AI can **perceive and act** in real time within its environment. This evolution is akin to giving AI "hands and feet" – enabling it to manipulate objects, navigate spaces, and directly assist with physical work. According to a World Economic Forum and BCG analysis, this shift to AI agents (both virtual and physical) marks a new era of near-autonomous systems that can perform complex tasks with minimal human intervention (Why should manufacturers embrace AI agents now? | World Economic Forum). In manufacturing settings, for instance, embodied AI agents (in the form of robots or smart machines) can make real-time decisions on the factory floor, moving industrial operations toward autonomy (Why should manufacturers embrace AI agents now? | World Economic Forum) (Why should manufacturers embrace AI agents now? | World Economic Forum). In everyday life, personal robots or AI-enhanced appliances could handle chores, and autonomous cars could chauffeur passengers – all examples of AI leaving the cloud to **live inside hardware that moves through our world**.

The Need for More Compute & On-Device Reasoning: This transformation greatly increases the demands on AI **computing and hardware**. Running a powerful AI model on a cloud server is one thing; running it on a robot or car in real time is another. Embodied AI requires significant **compute power on the edge** (within devices) and ultra-fast connectivity to the cloud. An autonomous vehicle, for example, must process camera and sensor data with low latency to make split-second decisions, using onboard AI chips. A humanoid robot guided by an AI brain may need to understand natural language, recognize objects, and plan movements – effectively an LLM and vision model working together on the device. Achieving this level of **AI reasoning in hardware** will demand new generations of high-performance, energy-efficient chips, larger memory bandwidth for handling data, and advanced power management (since many of these devices are mobile or untethered). In essence, as AI transitions from pure software to **software+hardware**, the hardware becomes a critical bottleneck to progress. This is why companies globally are now investing heavily in AI-specific semiconductors, edge computing solutions, and optimized devices to meet the requirements of embodied intelligence.

Notably, this hardware emphasis is reinforcing itself as a major trend. In a recent report, Morgan Stanley analysts describe the current wave as a **new technology cycle** driven by expanding use of GenAI and the need for more robust infrastructure (GenAI Revenue Growth and Profitability | Morgan Stanley). Enterprises are already planning to boost AI investments, with over 90% of companies expecting to increase spend on AI in the next three years (AI in the workplace: A report for 2025 | McKinsey) – and much of that spend will go into computing infrastructure to deploy AI widely. In short, the industry is evolving such that **the center of gravity of AI is shifting**: from cloud-hosted software accessible to specialized users, to **pervasive intelligent hardware** embedded in business operations, consumer products, and city infrastructure. This evolution underpins the investment themes in hardware and is the focus of the following sections.

Key Hardware Components Enabling AI Embodiment

Realizing embodied AI at scale will require an array of specialized hardware components. From semiconductor chips that provide the raw compute, to the sensors and batteries that enable autonomous machines, multiple layers of technology must come together. This section outlines the key hardware categories and their importance, along with examples of leading public companies in each area poised to benefit.

Compute & AI Chips

At the heart of every AI system is its compute engine. Today's AI models are computationally intensive, especially for tasks like deep learning, inference, and real-time decision-making. **Graphics Processing Units (GPUs)** have become the workhorse for AI training; companies like **NVIDIA** dominate this space with GPUs (e.g. the A100/H100) that are crucial for both training large models and running them at scale. NVIDIA's CEO has called generative AI's rise the "largest TAM expansion of software and hardware in decades," as demand for AI chips soars (AI's Trillion-Dollar Opportunity | Bain &

Company). Rival chipmakers **AMD** and **Intel** (through its Habana subsidiary) are also developing AI accelerators, while startups and new entrants design **ASICs** (Application-Specific ICs) tuned for AI workloads (e.g. chips by **Graphcore** in the UK or **SambaNova** in the US, albeit private companies).

Crucially, as AI spreads to edge devices, we see the emergence of **Neural Processing Units (NPUs)** and dedicated AI co-processors in phones, PCs, and IoT devices. Smartphone chip leaders like **Qualcomm** (Snapdragon platform) and **Apple** (with its Neural Engine in A-series/M-series chips) embed NPUs to enable on-device AI features (image recognition, voice assistants, etc.). In fact, by 2025 about **30% of new smartphones are expected to have on-board generative AI capabilities**, and nearly all PCs are projected to ship with some AI acceleration by 2028 (2025 semiconductor industry outlook | Deloitte Insights) (2025 semiconductor industry outlook | Deloitte Insights). These edge AI chips allow devices to run AI locally, which is vital for latency, privacy, and working offline.

In the data center, beyond GPUs, there are **Tensor Processing Units (TPUs)** like those developed by **Google** for its internal use, and **custom AI accelerators** from cloud players (Amazon's AWS Inferentia and Trainium chips, for example). There are also **DPUs (Data Processing Units)** and other specialized processors to offload networking and storage tasks for AI-heavy servers (companies like **Marvell** and **NVIDIA (Mellanox)** are active here). All these efforts point to a booming market for AI silicon. According to Morgan Stanley, spending on **AI-related hardware and networking is set to almost triple** from about $98 billion in 2024 to $276 billion in 2028 (GenAI Revenue Growth and Profitability | Morgan Stanley). Similarly, Bain & Company projects annual growth of **40–55%** in AI hardware/software TAM for the next several years (AI's Trillion-Dollar Opportunity | Bain & Company). The implication is clear: demand for compute chips to enable AI is skyrocketing. Companies leading in advanced chip design and manufacturing (such as Nvidia, AMD, **TSMC** in Taiwan for foundry, and **ASML** in Europe for chipmaking equipment) stand to benefit enormously as the world builds the brains for embodied AI.

High-Bandwidth Memory (HBM) & Advanced Memory

Feeding the beast of AI compute requires equally advanced memory technology. High-Bandwidth Memory, or **HBM**, is a specialized form of DRAM that is stacked in layers and located very close to the processor, enabling extremely fast data throughput. Modern AI accelerators (like Nvidia's H100 GPU) use HBM stacks to rapidly shuttle data in and out of the chip's cores, which is essential for training large neural networks. As models grow in size and complexity, memory bandwidth has become a critical limiter – and HBM provides a solution with bandwidth on the order of hundreds of GB/s per chip.

The **surge in AI demand has caused a boom in HBM**. Major memory makers **SK Hynix** and **Samsung Electronics** (both in South Korea) and **Micron Technology** (US) are the primary producers of HBM. SK Hynix, for instance, has reported unprecedented demand: it stated that its HBM3 chips (used in AI systems) were *completely sold out* for 2024 and almost fully booked for 2025 (Nvidia supplier SK Hynix says HBM chips almost sold out for 2025 | Reuters). The company's CEO noted the HBM market is expected to grow at ~60% annually for the mid-to-long term as AI model sizes increase (Nvidia supplier SK Hynix says HBM chips almost sold out for 2025 | Reuters). Micron similarly indicated its HBM lines are sold out well into next year (Nvidia supplier SK Hynix says HBM chips almost sold out for 2025 | Reuters). This scramble has led memory firms to accelerate innovation – e.g. Samsung recently announced a 12-high HBM3E stack to push capacity and speed even further. High-bandwidth and low-latency memory is **indispensable for AI**: whether in cloud servers or on-device (some edge AI chips use fast LPDDR or emerging memory tech), the ability to keep the AI "fed" with data is key to performance. Thus, memory suppliers are emerging as big winners of the embodied AI trend. In fact, industry forecasts suggest HBM will account for an expanding share of the total DRAM market by value (over 20% starting in 2024) as AI continues to drive demand (SK hynix's 41st Anniversary: Rise to AI Memory Leader) ([News] SK hynix CEO: Demand for Memory Chips to Remain Robust …).

Beyond HBM, other memory and storage technologies also see opportunity. **SSD makers** (like Samsung, SK Hynix, Western Digital) benefit because AI data centers require fast storage (often NVMe SSDs) to stream training data. Novel approaches like **Compute-in-Memory** (where processing happens directly in the memory arrays) are being researched to further speed up AI workloads – startups and labs (IBM, some Israeli firms) are exploring this to reduce data movement. In summary, investors should note that the **"memory" side of AI hardware is as crucial as the "compute" side**, and companies leading in high-speed memory are integral to the AI hardware ecosystem.

Networking & Data Center Infrastructure

When AI is deployed at scale, especially in cloud data centers or enterprise clusters, networking hardware becomes critically important. Training large AI models often involves **hundreds or thousands of chips working in parallel** across many servers – which means those chips need to communicate extremely fast. This has driven a wave of investment in **high-performance networking**: for example, **InfiniBand** and advanced Ethernet networks for AI clusters (Nvidia, through its Mellanox acquisition, is a leader in InfiniBand connectivity for AI supercomputers). Companies like **Arista Networks** and **Cisco** provide the switching gear for cloud providers to connect AI pods, and **Broadcom** and **Marvell** design the physical network interface chips and switch ASICs that handle massive bandwidth in data centers.

The data center itself is being re-architected for AI. Traditional server racks are being replaced or supplemented with **AI supercomputing infrastructure** – such as Nvidia's DGX systems or **HGX boards** that integrate multiple GPUs with high-bandwidth connections (NVLink, etc.). Hyperscale cloud firms (Amazon, Microsoft, Google, Meta) are building dedicated AI farms with custom interconnects. Indeed, we're witnessing an **AI data center build-out boom**: Morgan Stanley forecasts the **capex of hyperscalers (Amazon, Google, Microsoft, Meta)** to hit ~$300 billion in 2025, up roughly 50% year-over-year, largely driven by AI infrastructure investments (Morgan Stanley: Hyperscaler capex to reach $300bn in 2025 – DCD) (Morgan Stanley: Hyperscaler capex to reach $300bn in 2025 – DCD). These companies are not only buying equipment from others; many are also **developing their own hardware**. Google's TPU is one example of an in-house AI chip deployed in its data centers, and Amazon's Graviton and Trainium chips are similar efforts to optimize cloud AI workloads.

Networking within data centers isn't the only concern; **edge networking and 5G** are also enablers for embodied AI. Autonomous systems like drones or vehicles may rely on low-latency connections to the cloud (vehicle-to-infrastructure communication) for additional intelligence. Telecom operators upgrading to 5G and beyond could indirectly benefit by providing the connectivity tissue for AI-enabled devices (think of smart factories where machines are linked via private 5G networks). Additionally, **data center real estate and power** have become strategic assets due to AI – high-density AI servers consume vast power and cooling. This has shone a spotlight on data center REITs and power/cooling solution providers. Advanced cooling systems from companies such as **Schneider Electric, Munters and Vertiv** are increasingly in demand to manage the heat from AI hardware.

In short, the **plumbing of AI – the networks, interconnects, and data center facilities – is a major investment theme**. Goldman Sachs notes that if the first phase of the AI boom centered on chips (Nvidia's surge), the **second phase is now focusing on companies building AI infrastructure**, including "semiconductor designers and manufacturers, cloud providers, computer and network equipment makers, data center REITs, utilities, and security firms" (AI infrastructure stocks are poised to be the next phase of investment | Goldman Sachs). Many of these fall under networking and infrastructure. Look at **Cisco** (network switches), **Juniper Networks** (high-end routing for data centers), **Equinix** (data center colocation), or **Ciena** (optical interconnects) as other players that could see growth as the demand for AI capacity strains existing networks. We are effectively witnessing the construction of a new **digital backbone** purpose-built for AI workloads.

Battery & Power Technologies

Embodied AI often means **untethered devices** – robots, autonomous drones, electric vehicles – which in turn means reliance on advanced battery and power systems. If a humanoid robot is to operate for hours in a factory or if an autonomous vehicle is to drive passengers around all day, they need significant energy supply and efficient power management. Thus, **battery technology** is a key enabler for AI in mobility and robotics. Recent developments in battery chemistry (such as high-density lithium-ion cells, solid-state batteries in R&D by companies like QuantumScape, or fast-charging technologies) will directly impact how feasible and effective embodied AI systems can be. For example, **Tesla**, which is as much an AI company as an auto company, has invested deeply in battery innovation (from its 4680 cells to research into new chemistries) because better batteries extend the range and capabilities of its AI-driven products (both its self-driving EVs and its prototype humanoid robot, Tesla Optimus). In Asia, **Panasonic** (Japan), **BYD**, (China), **Samsung SDI** and **LG Energy Solution** (Korea) are giants in EV batteries that could benefit as demand rises not just from cars but also from autonomous platforms and electric robots.

Moreover, data centers themselves are power-hungry, and AI is *dramatically* increasing that hunger. Power and cooling are now as critical as processors in AI mega-clusters. Goldman Sachs projects that **global data center power demand will soar by 50% by 2027 (vs 2023) and by 165% by 2030**, largely due to AI workloads ([AI to drive 165% increase in data center power demand by 2030 | Goldman Sachs]) ([AI to drive 165% increase in data center power demand by 2030 | Goldman Sachs]). This creates opportunities for companies in the energy and power management sector: for instance, **Schneider Electric** and **Eaton** (which provide data center power infrastructure and uninterruptible power supplies) and utilities that specialize in supplying large-scale computing centers. Even traditional electric utilities could see a boost – as AI data center campuses (like those built by Microsoft or Google) draw more electricity, the grid investments and utility revenues associated with that are significant. Goldman analysts note that some utilities haven't yet seen their stock move on AI, but the **increased electricity demand** is very real ([AI infrastructure stocks are poised to be the next phase of investment | Goldman Sachs]).

There's also an intersection of AI and energy tech in the form of **smart grids and AI-managed power systems**. Companies providing smart energy solutions, grid battery storage, or AI-optimized power conditioning could see growth. Additionally, **power electronics** – e.g. efficient converters, chips made of silicon carbide (SiC) or gallium nitride (GaN) for high-efficiency power management in EVs and robots – are an important sub-sector. Firms like **Infineon** (Germany) and **ON Semiconductor** (US) are leaders in these segments and stand to gain as more electric, AI-enabled machines come to market.

In summary, for AI to step out of the cloud, it must sip (or gulp) electrons intelligently. Innovations in batteries (extending life, energy density), improvements in charging infrastructure (fast chargers, wireless charging for robots), and robust power infrastructure (both on-device and in data centers) will be critical. Companies solving the energy bottleneck – whether it's enabling a robot to run a full shift or powering a warehouse of AI servers – will be key beneficiaries in the embodied AI era.

Cybersecurity & AI Safety

As AI systems proliferate in hardware and potentially make autonomous decisions, ensuring their **security and safety** becomes paramount. We are not just talking about traditional cybersecurity (protecting servers from hacks) but also **AI-specific threats and safety challenges**. For example, consider a fleet of autonomous delivery robots or drones – if an attacker hijacks their AI, it could cause physical harm or disruption. Or consider factory robots making autonomous decisions; ensuring they are not fed corrupted data and that their actions remain within safe bounds is crucial. This opens the door for companies providing **AI-focused security solutions**: those that safeguard the integrity of AI models, protect the data pipelines feeding AI systems, and secure the hardware endpoints themselves.

Traditional cybersecurity firms are adapting their offerings to this new landscape. **Palo Alto Networks**, **Cisco**, **CrowdStrike**, and **Fortinet**, for instance, are all incorporating AI into their security products (both to defend against AI-aided cyber attacks and to protect AI systems). We might see specialized solutions for "AI model firewalling" or anomaly detection tailored to robotics and vehicles. Moreover, as AI hardware is deployed, there's risk of new attack surfaces (e.g., adversarial attacks on sensors, tampering with machine learning models). This is spurring a niche of **AI safety startups** focusing on things like validating AI model outputs, detecting AI model tampering, and ensuring compliance with regulations. While many such startups are private, large tech companies like **Microsoft** and **Google** (through Azure and Google Cloud) are also developing tools to help customers secure their AI deployments and to prevent issues like data leakage from AI models.

Another aspect is **chip-level security** for AI. Companies like **AMD** and **Intel** build secure enclaves and encryption into their chips (for instance, Nvidia's GPUs now come with features to isolate workloads and protect memory to prevent one AI tenant from interfering with another in shared cloud environments). As autonomous cars roll out, automotive cybersecurity (offered by **Harman**, owned by Samsung) becomes essential to prevent car hacking. And in critical areas like defense or infrastructure, ensuring AI-driven systems are fail-safe and cannot be manipulated is leading to new standards and certifications – benefitting firms with expertise in secure systems (think **Lockheed Martin** or **BAE Systems** on the defense side, which are working on AI-enabled but secure autonomous drones, etc.).

Overall, **trust will be a key currency for AI adoption**. Organizations will spend on tools and services that make their AI reliable and safe. Even consulting firms (Accenture, Deloitte) have growing practices in "AI risk and security" to advise enterprise and government clients. Goldman Sachs specifically pointed out that **security software providers are among the companies poised to gain in the next phase of AI investment**, as every new AI infrastructure deployment brings with it a need for protection (AI infrastructure stocks are poised to be the next phase of investment | Goldman Sachs) (AI infrastructure stocks are poised to be the next phase of investment | Goldman Sachs). In investment terms, this means cybersecurity companies that align offerings to the AI wave could see accelerated growth. It's a reminder that as we wire intelligence into physical systems, **safety nets must grow in tandem** – and that will be a significant spending priority (and opportunity) in the coming years.

Robotics & Autonomous Systems

Perhaps the most tangible manifestation of embodied AI is in **robotics and autonomous machines**. This spans a broad gamut: industrial robots in factories, service robots in retail or healthcare, humanoid robots that can generalize across tasks, autonomous vehicles and drones, and more. These are the ultimate "hardware bodies" for AI brains. Advances here combine improvements in AI software (like better vision, planning, and language models) with improvements in mechanical design, sensors, and actuators. The result is a new generation of robots that are more capable and flexible than the fixed, pre-programmed machines of the past.

Industries like manufacturing and logistics are already seeing a rapid uptick in robotics adoption. Global robot makers such as **ABB Ltd.** (Switzerland/Sweden), **Fanuc** and **Yaskawa** (Japan) are integrating AI to make their robots smarter – enabling tasks like quality inspection via machine vision or autonomous navigation of mobile robots in warehouses. These companies stand to gain as AI enhances the value proposition of automation. The Boston Consulting Group notes that AI agents (the intelligence controlling robots) can significantly amplify productivity, with early adopters in industry already achieving up to 14% cost savings and more autonomous operations (Why should manufacturers embrace AI agents now? | World Economic Forum). Essentially, smarter robots can do more, which drives more demand for robots.

On the consumer and service side, **autonomous vehicles (AVs)** are a high-profile example. Companies like **Tesla**, **Waymo (Alphabet)**, and **Toyota's Woven Planet** are pouring resources into self-driving AI. The vehicles themselves are hardware platforms laden with sensors (cameras, LiDAR, radar) and AI chips (for instance, Tesla's Full Self-Driving computer with Tesla-designed ASICs, or Waymo's adoption of custom AI hardware). As robo-taxis and autonomous trucks get closer to reality, auto OEMs and their suppliers (e.g., **NXP Semiconductors** and **Mobileye** for ADAS systems) will benefit. We already see regulatory approvals inching forward for autonomous taxis in certain cities, signaling a growing market. **Waymo's driverless cars**, for example, are operating ride-hailing services in Phoenix and San Francisco, showcasing the commercial potential of AI on wheels (Early Access to Waymo's Self-Driving Taxis. Here – YU Commentator).

Humanoid and general-purpose robots are another arena. While still nascent, prototypes like **Tesla's Optimus**, **Apptronik's Apollo**, **Figure AI's Helix** and **Agility Robotics' Digit** indicate a direction where a single AI-driven robot could perform multiple jobs (from warehouse stocking to delivery). If even a fraction of the vision for humanoid robots comes true, it could spark a massive hardware manufacturing industry (with parallels to how the automotive industry scaled). Key component makers – from **servo motor manufacturers** (e.g., Japan's Harmonic Drive, which makes precision gearboxes for robot joints) to sensor companies (**Ouster**, **Luminar** for LiDAR, or **Sony** which makes image sensors widely used in AI vision) – will ride this wave.

It's worth noting that **emerging markets and Asia are heavily investing in robotics and AI hardware**. China, for example, has national initiatives for AI and leads in some robotics deployments (with companies like **DJI** dominating drones, and others like **Hikvision** providing AI-powered surveillance hardware). These firms, though sometimes facing trade restrictions, underscore the global race in embodied AI. Meanwhile, companies in emerging economies are adopting robotics to leapfrog infrastructural gaps – e.g., autonomous drones for delivery in areas with poor roads, or AI-driven healthcare devices where doctors are scarce.

In sum, the robotics and autonomous systems sector represents the **apex of AI hardware integration** – where all the enabling technologies (chips, sensors, power systems, connectivity) come together in products that directly interact with the world. It is an exciting investment area: we are already seeing real-world traction (for instance, **warehouse automation robots** are a fast-growing market, benefitting firms like **Amazon (via Kiva Systems)** and **Shopify (via 6 River Systems)**, and industrial giant **Siemens** is embedding AI in factory equipment). As embodied AI becomes more capable, entire industries – from manufacturing to transportation to healthcare – could be transformed. The companies providing the robots or the critical components inside them are poised for significant growth in the next leg of the AI revolution.

(File:Atlas frontview 2013.jpg – Wikimedia Commons) *A new generation of **humanoid robots** and autonomous machines is emerging, enabled by AI. These robots combine advanced hardware (sensors, actuators, power systems) with AI "brains" to perform complex tasks in the physical world. Companies like Tesla, Hyundai (Boston Dynamics), and various startups are pushing the boundaries of robotics, aiming to deploy AI-driven robots in factories, warehouses, and everyday environments.* (Why should manufacturers embrace AI agents now? | World Economic Forum) (Why should manufacturers embrace AI agents now? | World Economic Forum)

The AI Data Center Buildout

The surge of AI adoption is triggering an **unprecedented buildout of data center capacity and semiconductor manufacturing** to support it. Just as the cloud boom a decade ago led to a massive expansion of hyperscale data centers, the **AI boom is now driving a new expansion cycle** – one that is even more compute-intensive.

On the **data center construction** front, hyperscalers (the big cloud firms) and even enterprise IT providers are investing heavily to create AI-optimized infrastructure. Morgan Stanley research indicates that combined capital expenditures of Amazon, Microsoft, Google, and Meta could reach **$300 billion in 2025**, up sharply from previous years, with much of the increase funnelled into AI-related infrastructure (Morgan Stanley: Hyperscaler capex to reach $300bn in 2025 – DCD) (Morgan Stanley: Hyperscaler capex to reach $300bn in 2025 – DCD). In 2024 alone, major cloud providers were expected to boost capex by ~36% year-on-year, largely thanks to AI and accelerated computing needs (Prepare for the Coming AI Chip Shortage | Bain & Company). These investments go into building gigantic GPU clusters, high-density server farms, and specialized facilities with the power and cooling to handle AI workloads. For example, Microsoft has been building out supercomputer-class clusters for OpenAI's models, Google is expanding its TPU pods, and Meta is refitting its data centers to be AI-friendly. The result is a wave of business for data center construction companies, engineering firms, and equipment suppliers (from cooling systems to generators). Regions with established tech hubs (Virginia's Data Center Alley, Silicon Valley, Ireland, Singapore, etc.) are seeing new AI capacity added, and even new regions (like the American Midwest, where some AI data center parks are planned) are joining the fray.

A critical piece of this buildout is **semiconductor manufacturing capacity**, especially at the leading edge. All those GPUs, TPUs, and AI accelerators need to be fabricated on cutting-edge process nodes (5nm, 4nm, 3nm, and soon 2nm). This hugely benefits the chip foundries – primarily **TSMC (Taiwan Semiconductor Manufacturing Co.)**, which currently produces the majority of advanced AI chips for companies like Nvidia, Apple, and AMD. TSMC has record orders and is running fabs at full tilt; it's also investing over $100 billion in the next few years to expand capacity (including new fabs in Taiwan, and in the US and Japan for diversification). **Samsung** in Korea is the other major advanced foundry and is likewise seeing strong demand for AI chip fabrication, while **Intel** is aiming to catch up in process tech and manufacture chips for others (including some AI chip startups) through its Intel Foundry Services. The **equipment makers** that supply these foundries are another indirect beneficiary: Netherlands-based **ASML**, which has a monopoly on EUV lithography machines required for the most advanced chips, is effectively selling every tool it can produce as chipmakers expand – AI demand is a key driver of their order books. Companies like **Applied Materials**, **Lam Research**, and **Tokyo Electron**, which provide deposition, etching, and other fab tools, also gain from the investment wave in new fab lines for AI chips.

We also see a push for specialized **packaging and assembly** technologies (often called "advanced packaging"). AI chips frequently use techniques like chiplet architectures, 2.5D integration (e.g., silicon interposers for HBM memory connectivity, known as CoWoS in TSMC's terminology (Chip-on-Wafer-on-Substrate (CoWoS) – TSMC – WikiChip)), and advanced cooling solutions. OSATs (outsourced semiconductor assembly and test companies) such as **ASE Technology** and **Amkor** are involved in packaging high-end AI chips with HBM stacks. Even government initiatives, like the U.S. CHIPS Act, target establishing domestic facilities for advanced packaging, with AI chips being a primary use case.

All told, the race to build out AI compute capacity is reshaping the tech infrastructure landscape. **Entire data centers are now being designed around AI**, whereas previously they might have been general-purpose. Goldman Sachs notes that many new high-density data centers are required, and even the global power grid will feel the impact (with data center power consumption tightening energy markets) (AI to drive 165% increase in data center power demand by 2030 |

Goldman Sachs) (AI to drive 165% increase in data center power demand by 2030 | Goldman Sachs). We are likely to see more collaborations between cloud providers and chipmakers – such as long-term supply agreements (e.g., Nvidia and cloud firms securing HBM memory supply from SK Hynix with multi-year commitments, or cloud players designing custom chips with foundry partners).

From an investment perspective, this buildout benefits a spectrum of companies: not just the obvious chip designers, but also construction firms, real estate trusts focused on data centers (e.g., **Digital Realty, Equinix**), electrical and cooling infrastructure firms (**Vertiv, Schneider Electric**), and the entire semiconductor supply chain (foundry and fab tool stocks have outperformed this year on AI demand). In addition, regions and governments are supportive – the EU, US, and China all see AI hardware as strategic, fueling further funding and possibly accelerating projects (like TSMC's fab in Arizona, or new state-backed AI compute clusters in Europe and Asia).

One striking aspect of this trend is scale: **data centers are getting huge**. Bain & Company pointed out that hyperscale AI installations might push data center sizes from today's ~100 MW of power capacity to **gigawatt-scale** in the not-too-distant future (AI's Trillion-Dollar Opportunity | Bain & Company). Imagine a single campus consuming as much power as a small city, entirely for AI computing – that gives a sense of the commitment being made. The buildout is not just about quantity but also technical sophistication: expect to hear more about **"AI supercomputers"** being stood up by various firms (for example, Meta's Research SuperCluster or Europe's planned Exascale AI systems). Each of these is a multi-billion dollar project feeding the ecosystem of hardware vendors.

In summary, the ongoing AI data center and chip fab expansion is a foundational enabler for the embodied AI revolution. It ensures that there will be enough "digital brains" to go around. For investors, many of these infrastructure plays are less hyped than, say, a pure-play AI software stock, but they represent picks-and-shovels with potentially more durable demand. As long as companies see competitive advantage in more AI compute (and current trends suggest they do, as larger models and more AI deployments continue unabated), this buildout cycle could run for several years, benefiting those firms building the physical backbone of AI.

(File:TSMC Fab5.JPG – Wikimedia Commons) *Semiconductor manufacturers and foundries are ramping up production to meet AI chip demand. Pictured: TSMC's Fab facility – companies like TSMC (Taiwan) and Samsung (Korea) are investing heavily in new fabs to produce advanced 3nm and 2nm chips, which are crucial for AI accelerators. Their equipment suppliers (ASML, Applied Materials) and various component providers benefit as global capacity expands (Nvidia supplier SK Hynix says HBM chips almost sold out for 2025 | Reuters) (Nvidia supplier SK Hynix says HBM chips almost sold out for 2025 | Reuters).*

Investment Themes & Opportunities

The transition to embodied AI presents a broad set of **investment themes**, especially in public markets. Companies that enable or supply the hardware components discussed are positioned to ride this secular trend. Below, we highlight key sectors and leading public companies (predominantly U.S.-based, with notable international peers) that stand to benefit in the next 1–3 years, along with the magnitude of the opportunity as projected by industry research.

● **AI Compute Chips:** The **semiconductor industry** is at the forefront of AI's investment boom. GPU leader **NVIDIA (NVDA)** has been emblematic – its stock surged as investors realized its chips power practically every AI data center. But beyond Nvidia, watch **AMD (AMD)** (gaining traction with its MI series accelerators) and **Intel (INTC)** (leveraging its CPUs plus specialized chips like Gaudi and Movidius). **TSMC (TSM)**, while not a household name to all investors, is crucial as the contract manufacturer for most high-end AI chips – it captures value from *all* chip designers' growth. Similarly, **ASML (ASML)** in Europe, which sells the lithography machines needed for advanced chip fabrication, has an almost toll-booth-like position on the AI chip boom. Investment banks have noted that the **AI semiconductor market is set to grow explosively** – Morgan Stanley forecasts GenAI-related semiconductor spending will rise from $115 billion in 2024 to **$280 billion in 2028** (GenAI Revenue Growth and Profitability | Morgan Stanley). That rising tide could lift many chip stocks. Other niche players: **Marvell (MRVL)** and **Broadcom (AVGO)** (networking and accelerator chips for data centers), **Qualcomm (QCOM)** (for edge AI in devices), **Texas Instruments (TXN)** (analog and embedded chips that will go into sensors and power management for AI hardware).

● **Memory and Storage:** AI's appetite for memory is a boon for **DRAM and NAND manufacturers**. **SK Hynix and Samsung** (in Asia) and **Micron Technology (MU)** (US) form an oligopoly in advanced memory. All are ramping up HBM production and fast DDR5 memory for servers. As cited, HBM could exceed 20% of DRAM market value starting in 2024 due to AI demand ([News] SK hynix CEO: Demand for Memory Chips to Remain Robust …). These firms have already seen financial upside: SK Hynix reported record revenues and a return to profit driven by AI memory sales (SK hynix hits record profit powered by AI memory demand). Investors can also consider **Kioxia (Japan, planning IPO)** or **Western Digital (WDC)** for NAND flash used in AI training storage. Additionally, **Seagate (STX)** sees demand for high-capacity HDDs in AI archives. Goldman Sachs research suggests that **hardware (including memory) spend in AI may rival or even surpass software spend** in coming years as companies race to build capacity (GenAI Revenue Growth and Profitability | Morgan Stanley) (GenAI Revenue Growth and Profitability | Morgan Stanley), which underscores the opportunity in this segment.

● **Cloud/Hyperscaler Ecosystem:** The big cloud operators – **Amazon (AMZN)**, **Microsoft (MSFT)**, **Alphabet (GOOGL)** – while spending heavily (which is a cost), are also turning AI into revenue growth. They are selling AI cloud services at premium prices and will likely see higher utilization of their data centers. So owning hyperscalers can be an indirect hardware play (they benefit from the AI services enabled by their infrastructure investment). However, more pure-play opportunities exist in companies supplying them: **Arista Networks (ANET)** for high-performance switches (widely used by cloud giants for AI clusters), **Equinix (EQIX)** and **Digital Realty (DLR)** for data center space leased to firms expanding AI capacity. **Lenovo** and **Super Micro Computer (SMCI)**, which build AI servers and appliances, have seen upticks in orders – SMCI, for instance, has grown rapidly by specializing in GPU server solutions for AI.

● **Networking & Telecom:** As previously noted, **Broadcom (AVGO)** (also a major supplier of AI networking chips and custom ASICs – it even supplies Google's TPU interconnect and AWS's custom silicon needs) is benefiting. **Cisco (CSCO)** is integrating AI features in its routers and aiming to capture the data center switching demand from AI, though Arista has been winning share in that niche. Telecom companies might seem peripheral, but **Verizon, AT&T** etc. could see enterprise 5G opportunities for AI (though that's longer-term). More directly, **Ciena (CIEN)** is seeing cloud providers upgrade backbone capacity to handle massive AI data flows between data centers.

● **Robotics & Automation Companies: Tesla (TSLA)** deserves mention not just for its autonomous driving (which effectively makes every Tesla car an AI hardware product) but also for the potential of its humanoid robot project. While Optimus is early, Tesla's AI expertise and vertical integration (including its own AI chips, Dojo supercomputer, and battery tech) position it uniquely if humanoids become a market. **Autonomous vehicle tech** firms like **Mobileye (MBLY)** (computer vision systems) and **Luminar (LAZR)** (LiDAR for self-driving) offer picks-and-shovels for the AV revolution. In industrial robotics, **ABB (ABB)** and **Fanuc (FANUY)** are publicly traded stalwarts – they are incorporating more AI into their product lines (ABB has AI software for robot picking, Fanuc has AI for predictive maintenance). **Intuitive Surgical (ISRG)** is another interesting one – while known for surgical robots, it's increasingly using AI to improve those systems and could expand into other medical automation. Also, **John Deere (DE)** – not often thought of in AI discussions – is embedding AI in farm equipment (autonomous tractors, smart combines), representing how even traditional equipment makers can see AI-driven growth and thus hardware sales.

● **Batteries & Electric Mobility:** On the battery side, pure-play cell manufacturers like **CATL (China)** and **Panasonic (PCRFY)** (supplier to Tesla) will indirectly benefit from more robots and EVs. **QuantumScape (QS)** and **SolidPower (SLDP)** are speculative plays on solid-state batteries that, if successful, could revolutionize energy density for AI devices (though their timelines are longer). **ChargePoint (CHPT)** and **ABB** (for EV chargers) could see an uptick as autonomous EV fleets expand and need widespread charging infrastructure. Additionally, **energy storage companies** (like **Fluence (FLNC)** or **Tesla's energy division**) might get a boost because large AI data centers often pair with on-site battery storage to manage power (for cost and backup reasons).

● **Security & AI Software Leaders:** While hardware is the focus, it's worth noting **AI software leaders** will also gain – and some have hardware-adjacent offerings. For instance, **Palantir (PLTR)** is developing AI platforms for military and industry and may bundle them with edge hardware for deployments. **Snowflake (SNOW)** and **Databricks (upcoming IPO)** are integrating with on-premises AI hardware for big enterprises. These aren't hardware plays per se, but they ride the same trend of AI proliferation. On security, as mentioned, companies like **CrowdStrike (CRWD)**, **Palo Alto Networks (PANW)**, **Zscaler (ZS)** are all touting AI in their products and will be instrumental in safeguarding AI infrastructures – as AI deployments grow, their addressable market grows.

To quantify the **magnitude of the opportunity**, consider some top-down estimates: Bain & Company projects the **market for AI-related products and services will approach $1 trillion by 2027** (AI's Trillion-Dollar Opportunity | Bain & Company). This includes software and hardware – but notably, Bain expects roughly half of this value to be in hardware (chips, devices, infrastructure) as opposed to purely software services. Consulting firm McKinsey likewise notes multi-trillion dollar economic impacts, which will translate into hundreds of billions in technology spend. And Goldman Sachs analysts have highlighted that the **"next phase" of the AI trade is moving toward infrastructure builders and away from just one or two big winners** (AI infrastructure stocks are poised to be the next phase of investment | Goldman Sachs) (AI infrastructure stocks are poised to be the next phase of investment | Goldman Sachs). In practical terms, this means a diversification of investment opportunities: instead of only the AI model owners (OpenAI, etc.) benefiting, we'll see broad-based gains for those selling the enabling tech – from chip fabs to sensor makers to cloud landlords.

Finally, it's important to recognize a geographic spread in opportunities. The U.S. has the majority of leading AI hardware companies (chips and cloud). But **Europe** has crown jewels like **ASML**, **STMicroelectronics (STM)** (which makes specialized chips and sensors, e.g., machine vision sensors, and power semiconductors), and **ABB** (robotics). **Asia** is home to TSMC, Samsung, SK Hynix, Sony (dominant in image sensors for AI cameras), and emerging Chinese AI chip designers (though U.S. export controls have complicated investing there). **Emerging markets** may not have many big AI hardware manufacturers yet, but they will be massive adopters – for instance, India's Reliance Jio is investing in AI infrastructure to serve its telecom and retail network, and could favor certain vendors. Also, Taiwan's and South Korea's economies are heavily leveraged to the AI chip boom through TSMC, Samsung, Hynix, etc., meaning their markets could outperform if AI demand stays strong.

In summary, the investment landscape for embodied AI is rich and spans multiple sectors. A useful strategy is to align portfolios with the core components: **compute, memory, connectivity, power, and automation**. Each of these has a set of leading firms with robust moats and significant capacity to scale. And as the next section concludes, we are still early in this hardware cycle, with an inflection point upon us that could sustain a multi-year growth trajectory for these companies.

(File:BalticServers data center.jpg – Wikimedia Commons) *Hyperscale **data centers** are being refitted as AI supercomputers. This is driving demand for high-performance servers and networking gear. Here, racks of server nodes with specialized AI accelerators glow with status lights. Companies like Arista Networks (switches) and Broadcom (networking chips) supply the critical connectivity that links thousands of these machines into powerful AI clusters (Morgan Stanley: Hyperscaler capex to reach $300bn in 2025 – DCD) (Morgan Stanley: Hyperscaler capex to reach $300bn in 2025 – DCD).*

Conclusion: The Inflection Point for AI as an Industry

The rise of embodied AI signals an **inflection point** where artificial intelligence moves from a niche capability to a pervasive general-purpose technology embedded throughout the economy. In prior tech revolutions, we saw similar inflections – the moment PCs became affordable for every business, or when the internet shifted from dial-up to broadband – unleashing a new wave of productivity and wealth creation. We are at such a moment for AI. By bringing AI into hardware, the technology is poised to break out of the digital realm and start transforming the physical world in earnest. This could unlock enormous economic value: enterprise efficiency gains, new consumer experiences, and even helping address labor shortages in aging societies through automation. McKinsey researchers liken today's AI trajectory to the early internet and note that **the risk for business leaders now is not thinking too big, but rather too small** in terms of AI's potential (AI in the workplace: A report for 2025 | McKinsey). In other words, those who ambitiously invest in and embrace AI (including its hardware embodiments) stand to lead, while laggards may quickly fall behind in competitiveness.

From an industry standpoint, **AI is transitioning from a period of research and pilot projects to one of scale deployment**. This transition will benefit not only the high-profile AI model creators, but a wide ecosystem of suppliers and integrators – many of which we identified in this report. The next 1–3 years will likely see rapid iterations in AI hardware: new chip launches, prototypes of AI robots moving to commercial products, first fully self-driving fleets in cities, etc. Each milestone will reinforce the virtuous cycle: successful use cases will spur further investment in underlying infrastructure. Already we see positive feedback loops, such as AI datacenters enabling the training of even more powerful models, which then open new applications that require more hardware to deploy (for instance, advanced generative AI models giving rise to AI copilots in software, which then demand edge AI chips in laptops for better performance). It's a self-reinforcing cycle of growth.

The **industries best positioned** to capitalize on this transition are those that combine domain expertise with AI adoption. Manufacturing and logistics stand to gain hugely from robotics and AI-driven optimization. Healthcare could see a proliferation of AI-powered diagnostic devices and assistive robots. Transportation will be reshaped by autonomy. Tech infrastructure firms (semiconductors, cloud) will enjoy robust demand as everyone else upgrades to "AI-ready" capabilities. Even sectors like construction or agriculture – traditionally not associated with cutting-edge IT – could be transformed by AI-driven machinery (e.g., autonomous construction equipment, smart drones for crop management). This broad impact underscores why consulting firms like McKinsey and BCG estimate AI (and automation) could raise global GDP by multiple percentage points and add *trillions* of dollars in value annually across sectors (Economic potential of generative AI | McKinsey) (AI's Trillion-Dollar Opportunity | Bain & Company).

For investors and stakeholders, the **focus now should be on execution and scaling**. The technologies are largely known; the race is in implementing them effectively. Companies that have strong footing in the hardware aspects – be it via intellectual property (patents, trade secrets in chip design or robotics) or via manufacturing capacity or supply chain control – have an edge in this scaling phase. It's telling that even software-centric firms (like Google and Amazon) felt the need to develop custom hardware to secure their AI future. Hardware has become strategic. As a result, we may see more M&A activity where big players acquire smaller chip designers or robotics firms to integrate vertically. We may also see partnerships between, say, automakers and tech companies to marry hardware expertise with AI know-how (as seen in deals like GM and Honda with Cruise, or Toyota investing in autonomous tech startups).

In conclusion, the transition from cloud-based AI to **embodied AI** is ushering in the next major growth cycle for the tech industry – one that could be even larger in scope than the last, because it touches every tangible part of the economy. The opportunity for investment is correspondingly large. **AI hardware spending is set to rival or exceed software spending** in this cycle, flipping the script of the past decade. Public companies in the U.S., Europe, and Asia that are building the "arms and legs" of AI – the chips, the machines, the infrastructure – are on the cusp of significant growth. As a final thought, it's worth remembering how quickly internet-era leaders emerged in the 2000s and mobile-era leaders in the 2010s. The mid-2020s may similarly see the rise of new titans of AI hardware. Those investing now, with a clear strategic vision (and not shying away from the bold scale of the opportunity), will be in the best position to reap the rewards of AI's next evolution. In the words of Jensen Huang, *"Generative AI is just the beginning – the next step is when AI gets out of the computer and into the world, and that's when things truly get interesting."* The journey from software to hardware in AI is underway, and its destination could fundamentally reshape our global economy.

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